

Summary

Full-Stack AI Engineer — 3+ years building scalable real-time systems and secure multi-tenant backends. Skilled in TypeScript, Postgres RLS, WebRTC, AI-powered features, Next.js, and AWS. Designed and shipped horizontally scalable RTC and HLS pipelines, improved backend throughput 45%, and supported 3k+ concurrent users with median latency <200ms (validated via load tests)

Work Experience

- Full-Stack Developer — 3Zero Digital**
Chattogram, Bangladesh | Jan 2025 - Dec 2025
 - Designed a worker-pool architecture with batched writes, boosting throughput 45% and cutting latency 50% while maintaining peak-load stability.
 - Deployed Agentic RAG on MCP servers, automating AI triage and reducing support tickets by ~25%.
 - Set up CI/CD pipelines with GitHub Actions and Docker for automated testing and deployment.
 - Added structured logging, tracing, and rate limiting, cutting MTTR 35% and raising system reliability.
 - Refactored the monolith into microservices, improving deployment flexibility.
 - Delivered conversion-focused UI with Next.js + shadcn/ui, boosting key flow conversion 12% through A/B testing.
- Freelance Full-Stack Engineer**
Remote | Mar 2023 - Dec 2024
 - Architected and delivered full-stack systems for diverse clients, building secure backends, modern frontends, and optimized DB schemas. Focused on authentication, realtime APIs, and performance tuning to improve scalability and reliability.

Selected PROJECTS

- Social Match-Making Video Chat Reels (Monorepo).....** [Live Link](#) | [Code](#)
 - Engineered a horizontally scalable RTC system with WebRTC, NATS, Socket.io, and Redis; integrated a NATS-PostgreSQL pipeline achieving sub-100ms message acknowledgment.
 - Architected an AWS ECS + FFmpeg HLS pipeline for seamless video streaming; leveraged Neo4j graph database to power relationship-based discovery and matching.
 - Implemented JWT session authentication and Redis-backed rate limiting; optimized performance with batched DB writes and integrated OpenAI SDK chatbot.
- Multi-Vendor E-Commerce (Monolithic).....** [Live Link](#) | [Code](#)
 - Built a secure Next.js 16 platform using Drizzle ORM and PostgreSQL RLS to ensure strict multi-tenant data isolation at the database level.
 - Guaranteed data integrity by implementing Database Transactions for atomic order processing, inventory sync, and secure Cryptomus crypto payments.
 - Engineered separate Dashboards for vendors, admin with ABAC (CASL), real-time delivery tracking and multi-channel notifications, improving efficiency by 20%.

Core SKILLS

Languages & Frameworks: TypeScript, Python, Next.js, React-Native, NestJS, Fastify, GraphQL.
Data & DB: PostgreSQL (RLS), Drizzle ORM, Redis, MongoDB, Neo4j, Qdrant.
Infra & Messaging: Docker, AWS (VPC, ECS, EC2, Lambda, SNS, SQS, S3, CloudFront, Bedrock), Terraform, NATS, RabbitMQ, NGINX, Cloudflare (Domain, R2, Workers, CDN, AI).
Auth & Security: JWT/JWKS, OAuth, RBAC, ABAC, Zod, CASL, Passport.js, BetterAuth.
Realtime & Streaming: WebRTC, Socket.io, HLS, FFmpeg.
AI / Tools: Agentic RAG, MCP Servers, A2A, LangGraph, n8n, OpenAI SDK.
DevOps / Observability: Linux, GitHub Actions, Prometheus/CloudWatch, Structured Logging, CI/CD.
Methodologies: Event-Driven Architecture, SOLID, KISS, Agile.

EDUCATION

- Master of Science in Computer Science & Engineering**, International Islamic University Chittagong — 2026
 - Thesis:** Leakage-Aware Multi-Class Detection of AI-Generated Phishing Emails Using Stylometric Features and Stacked Ensemble Learning.
- Bachelor of Science in Computer Science & Engineering**, Port City International University — 2023
 - Thesis:** Gender Predictions of Bengali Names: An Ensemble and BERT-based Approach.